RavenTag

Authenticate. Verify. Trust.

The open-source NFC authentication framework
that links physical products to the blockchain.

Powered by NTAG 424 DNA + Ravencoin

- Hardware-secured NFC chip
- Ravencoin blockchain proof
- Brand-sovereign key control

Counterfeiting is a $500B+ problem

Every year brands lose billions, consumers are deceived,
and existing solutions lock you into expensive, fragile systems.

Expensive SaaS

Third-party platforms charge per-scan fees and subscriptions that scale against you.

Keys held by others

Your authentication keys live on someone else's server. One breach exposes everything.

Vendor lock-in

Proprietary ecosystems trap your brand. Switching means re-tagging your entire line.

No on-chain proof

Centralised databases can be altered. No public, permanent, tamper-proof record exists.

RavenTag: trustless product authentication

Three technologies. One tap. Zero trust required.

NTAG 424 DNA

NXP hardware chip embedded in the product. Generates a cryptographically unique URL on every tap. AES-128 keys stored in tamper-proof silicon, impossible to clone.

Ravencoin Blockchain

Each item is a unique on-chain token. The IPFS metadata binds the serial number to the chip fingerprint. Public, permanent, censorship-resistant proof.

Brand Key Sovereignty

AES master key lives only on the brand's own server. Keys are derived per-chip server-side. Nothing secret ever leaves your infrastructure.

Three steps from factory to consumer

01

Brand Issues Token

Brand creates a unique Ravencoin token for each item:
FASHIONX/BAG01#SN0001

IPFS metadata stores:
nfc_pub_id = SHA-256(UID + salt)

>

02

Chip is Programmed

Android app calls POST /api/brand/derive-chip-key.
Backend derives 3 AES-128 keys from BRAND_MASTER_KEY.
Keys written to NTAG 424 DNA chip via NFC.

>

03

Consumer Verifies

Consumer taps product with any NFC phone.
Brand server verifies SUN MAC + blockchain record.
Instant result: AUTHENTIC / REVOKED / INVALID

Four independent security layers

All four must be simultaneously compromised to forge a valid tag. No known attack vector achieves this.

L1

Ravencoin Consensus

Sub-asset tokens can only be issued by the parent asset holder. Every full node enforces this independently.

L2

nfc_pub_id Binding

SHA-256(UID + BRAND_SALT) ties each chip to the brand. Requires BRAND_SALT, never stored on-chain or public.

L3

AES Key Derivation

AES-ECB(BRAND_MASTER_KEY, [slot + UID]) per chip. Requires BRAND_MASTER_KEY, never leaves brand server.

L4

NTAG 424 DNA Silicon

Keys stored in write-protected secure memory. Non-clonable NXP hardware, physically impossible to duplicate.

From handbag to high jewellery

Each bag gets its own blockchain token

FASHIONX/BAG01#SN0001, immutable, public, permanent.

Chip embedded invisibly

NTAG 424 DNA hidden in lining, strap, or hardware fitting.

Instant buyer verification

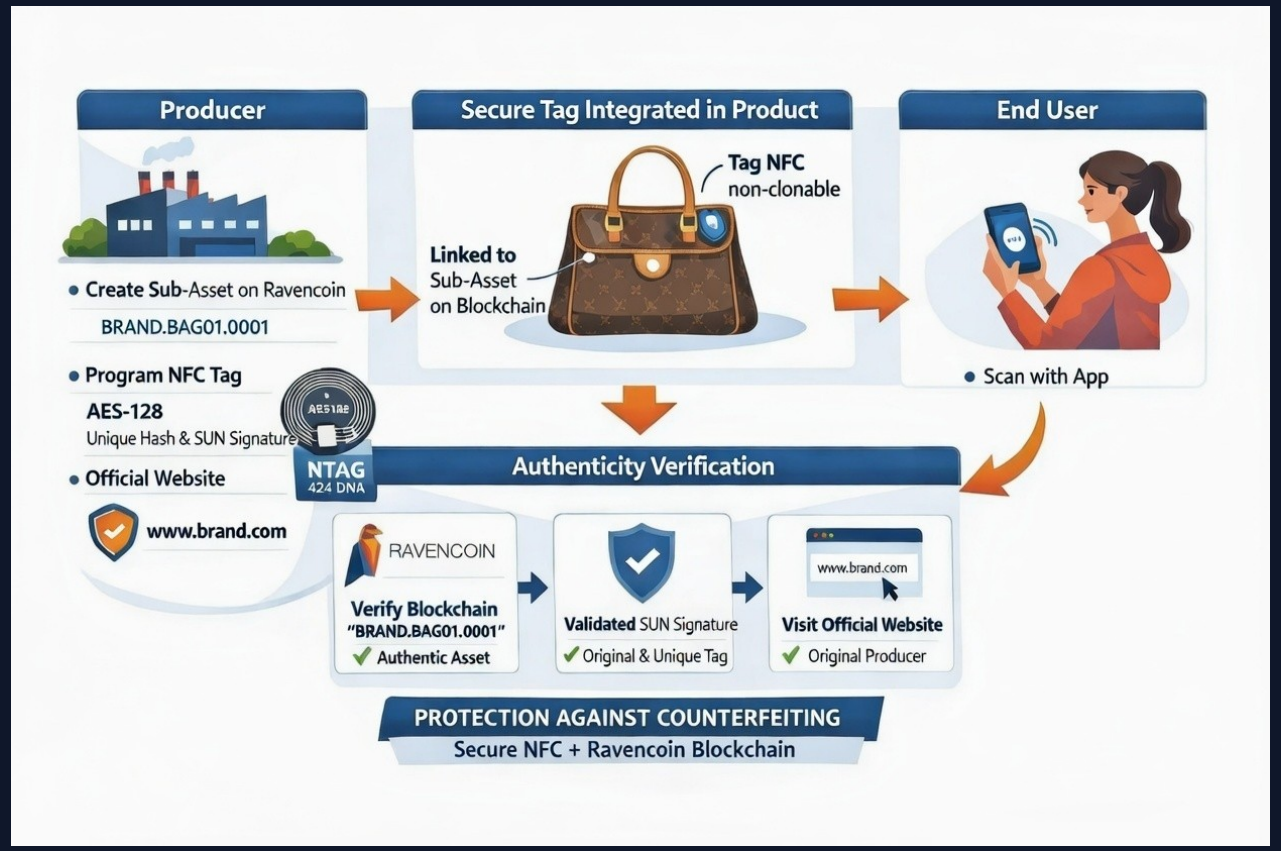
Tap phone to product. Authentic in under 2 seconds.

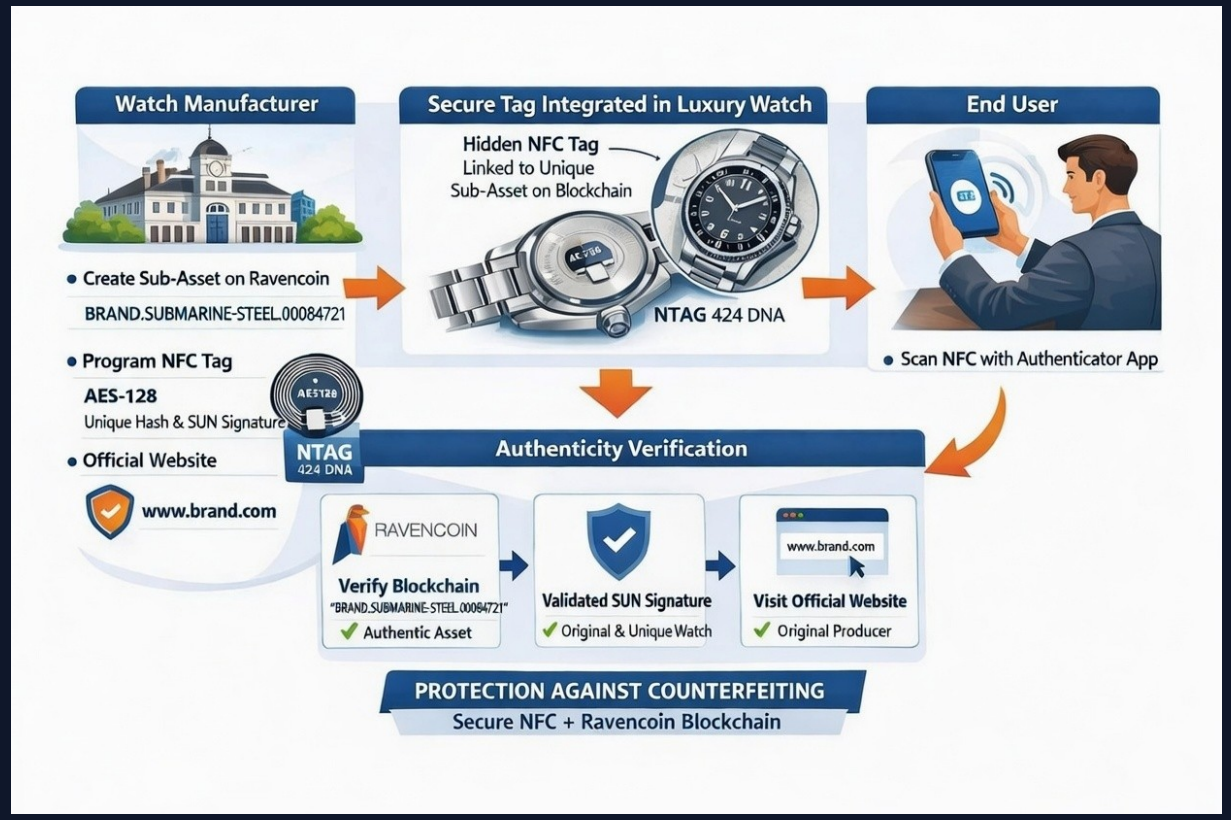
Revocation on theft or loss

Mark any item as stolen via API. Verification instantly fails.

Resale market trust

Second-hand buyers verify provenance without contacting brand.





Luxury watches deserve unbreakable proof

Hidden chip, zero design compromise

NFC tag fitted inside caseback or bezel. No QR codes, no holograms.

Serial number on-chain

BRAND.SUBMARINER.00084721 as a unique Ravencoin token.

Service history immutability

Brand annotates IPFS metadata with service records.

Anti-cloning guarantee

NTAG 424 DNA SUN MAC is unique per tap. Replays are detected.

Works without app install

Verification URL opens in any mobile browser via NFC.

Why Ravencoin?

Native asset protocol

Root assets, sub-assets, and unique tokens are first-class protocol citizens. No smart contracts, no bytecode vulnerabilities.

KAWPOW Proof-of-Work

ASIC-resistant mining keeps the network decentralised and censorship-resistant. No corporate control.

No ICO, no premine

Ravencoin launched in 2018 with zero pre-allocated coins. Fully community-owned and MIT licensed.

5 RVN per unique token

Per-item serialisation is economically viable even at millions of units. Fractions of a cent per product.

Consensus-enforced ownership

Every full node verifies that sub-assets belong to the parent asset holder. No central authority needed.

Open source blockchain

No vendor dependency at the blockchain layer. Brands run their own Ravencoin node if needed.

Ravencoin

ravencoin.org

Bitcoin fork built
for asset issuance.
No ICO. No premine.
Community owned.

The brand case for RavenTag

Feature	Traditional SaaS	RavenTag
Annual cost	$5,000 to $100,000+	~$15/year (domain only)
Keys stored by	Third-party vendor	Brand's own server
On-chain proof	No	Yes, Ravencoin token
Vendor dependency	Total	Zero, open source
Revocation	Slow / manual	Instant via API
White-label consumer app	Paid add-on	Included, customize freely
Operator role separation	Rarely	Admin + Operator keys
Self-hosted	No	Yes, Docker + nginx

Two apps for every role in the chain

Brand Manager App

For operators and brand teams

- Issue Ravencoin assets (root, sub-asset, unique token)
- Derive chip keys via backend API
- Program NTAG 424 DNA chips via NFC
- HD wallet: BIP44/BIP39, local signing
- Revoke, burn, transfer asset lifecycle
- Multi-language: EN, IT, FR, DE, ES

./gradlew assembleBrandRelease

Consumer Verify App

For end buyers, fully white-label

- Scan NFC tag and verify authenticity instantly
- Shows AUTHENTIC / REVOKED / INVALID result
- Built-in Ravencoin wallet
- Fully customisable: logo, color, brand name
- No app install needed for basic web verify
- Multi-language: EN, IT, FR, DE, ES

./gradlew assembleConsumerRelease

Production-ready in one afternoon

Total infrastructure cost: ~$10-15 USD/year (domain only). Everything else is permanently free.

Oracle Cloud ARM

Backend: Node.js + SQLite
4 OCPU, 24 GB RAM
Always Free, never shuts down

Free

Vercel

Frontend: Next.js 14
Auto-deploy on git push
Global CDN included

Free

Cloudflare

DNS + CDN + SSL
DDoS protection
Auto HTTPS certificates

Free

Domain

Cloudflare Registrar
or Porkbun
.xyz from $1/year

$1-15/yr

docker compose up -d >> git push >> done.

Fully open. Zero vendor lock-in.

Apache 2.0 License

Fork freely. Modify, redistribute, embed in commercial products. Attribution required per clause 4e.

Self-hosted

Deploy on your own infrastructure. No dependency on RavenTag services for core operation.

Protocol: RTP-1

The RavenTag Protocol v1 spec is public. Any developer can implement a compatible client or server.

Brand independence

Your AES keys, your blockchain assets, your server. RavenTag can shut down and your system keeps running.

White-label apps

Both Android apps are open source. Customise branding, colors, and endpoints for your product line.

CI/CD included

GitHub Actions workflow builds backend, frontend, and Android APKs on every push. Ready for production.

RavenTag

Start authenticating your products today.

Open source. Self-hosted. Blockchain-verified.
The only anti-counterfeiting system your brand fully controls.

Source code	github.com/ALENOC/RavenTag
Download APK	github.com/ALENOC/RavenTag/releases
Documentation	raventag.io

Free forever

Oracle Cloud Always Free
Vercel Free Tier
Cloudflare Free Plan

Only pay for your domain
~$10-15 / year